Building a large tile array in free space: what our simulations say, and what they do not

A report on the ORLO-Zig assembly-strategy study, for readers who did not take part

ORLO-Zig project

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Executive summary

The question. A robot must build a ten-by-ten array of 100 tiles in free space, with no gravity, no ground and nothing to push against. Every move of the robot and of every tile pushes back on the part that is already built. The growing structure therefore turns while it is assembled. We asked: *does the order in which the tiles are laid change how much the structure turns and how long the motion takes, and what should we test next?*

The model. A rigid-body, isolated, prescribed-motion model. The system starts at rest with zero linear and zero angular momentum. Robot, gantry and tile motions are prescribed. The model computes how the structure must turn to keep the total momentum at zero.

What we found (all numbers are model results; tag [V] means "reproducible from retained files"):

- The standard *snake* route (back and forth along the rows, "snake-baseline") takes 303.8 s of motion. The structure has a sampled peak turn of 61.7° from its start orientation (a lower bound: the true peak can be larger), and it ends 61.7° away from it.
- The *outward spiral* takes 309.1 s. It ends closer to the start orientation (46.4°). But its sampled peak is pinned at the largest value the angle can take, about 180°. The true peak is not resolved.
- Neither route is better on all measures. The difference in motion time is only 5.3 s (derived).
- The snake passed all model checks in 10 of 10 stress variants (changed masses, inertias and centre of mass). The outward spiral passed 2 of 10; the other 8 gave an unknown result.
- A bounded search of 12 full simulations found 2 further routes. None beat the snake in its own comparison group.

What we do not know. The model does not show that any route can be built. Whether the gantry can reach every tile, whether the magazine can hold mixed-orientation tiles, whether the rail and drive can do the turns, how long one seam takes to join, and how much joint load is safe are all unknown ([U]). Seam joining may dominate the real build time: if every one of the 180 seams took an assumed 600 s, motion plus joining would take 30.1 h in total and the motion would be 0.28 % of that (assumed [A], derived).

Recommendation. Carry the snake forward as the *reference* route. Do not call it a winner. Keep the outward spiral as a conditional alternative. Before any choice between them, run a small physical test. We propose a bench test of seam joining first, and a gantry and magazine mock-up second (Section Next validation experiment).

Claim limits that apply to every statement in this report. Captions and text refer to them by their codes L1 to L8.

- **L1.** Rigid, isolated, prescribed-motion model only.
- **L2.** Physical feasibility (reach, clearance, magazine indexing, rail contact, drive, seam capacity, joint safety) is unknown.
- **L3.** Motion time is not completion time: serialized seam joining dominates (for example 30 h at 600 s per seam versus about 5 min of motion).
- **L4.** Attitude and load peaks are sampled lower bounds; the spiral peak is pinned near 180 deg.
- **L5.** No global optimality.
- **L6.** SQ100 and SHAPE100 are not ranked against each other.
- **L7.** The robot marker in movies is not CAD.
- **L8.** The route-library and search results were computed with an earlier version of the evaluation code. The code has changed since then, and the retained runs were not re-computed with the current code. The nominal snake was re-run with the current code and gave the same result.

1 The problem in plain words

1.1 A free-floating build

On the ground, a robot that places a tile can lean on the floor. In orbit it cannot. We model a structure with nothing outside it: no gravity, no drag, no external force and no external torque. Two quantities are then conserved for the whole system (robot, gantry, magazine and built structure together): the total linear momentum and the total angular momentum. Both start at zero, because everything starts at rest.

Zero total momentum does not mean that nothing moves. It means that motion of one part must be matched by motion of the others. The centre of mass of the whole system stays fixed in space. The *orientation* of the structure is not fixed, and that is the quantity we study.

1.2 Why the structure rotates

Three effects turn the structure.

1. **The robot drives along the rail.** The robot is a large part of the system mass, and it is not at the system's centre of mass. When it accelerates along the structure, the structure must move the other way to keep the total momentum at zero. The robot's motion also carries angular momentum about the system's centre of mass, and the structure must take up the opposite share by rotating.
2. **The robot turns.** At each turn tile the whole robot body turns by a right angle. Its spin has angular momentum. The rest of the system takes up the opposite angular momentum and rotates.
3. **The mass moves.** Tiles leave the tall magazine stack and join the structure. The centre of mass of the structure moves, and so does its moment of inertia. The same push then gives a different rotation.

The result depends on the *order* of the moves. Turns in the same direction in a row can add up. The first turn of the outward spiral rotates the structure by 88.5°, against 20.2° for the snake. The outward spiral reaches 1124° of unwrapped yaw at the end of the build. That large number shows repeated rotation, not cancellation (it is an accounting of the total turn, not a physical angle of the final structure).

1.3 How we measure "how much it turned"

Attitude error is the smallest rotation angle that takes the structure from its start orientation to its current orientation. It runs from zero to 180° at most (the largest possible value is exactly half a turn). We report two numbers: the *peak* over the whole build, and the value at the *end* of the motion. A route with a small final value can still have swung through a large angle in between. We also report *motion time*, the modelled time that the robot, gantry and placements take. Motion time is not the time to finish the build (see the glossary and Section Results).

1.4 Two strategies, A1 and A2

The study started from two ways to organise the work, called A1 and A2. In **A2**, one robot of dry mass 85 kg carries all the tiles, drives forward along the path and places each tile (system mass 135 kg). In **A1**, an anchor robot stays at the first tile with the magazine. A lighter crawler carries tiles out along the path in ten sorties and then returns to the anchor. A2 has no return trips. A1 has many more starts and stops.

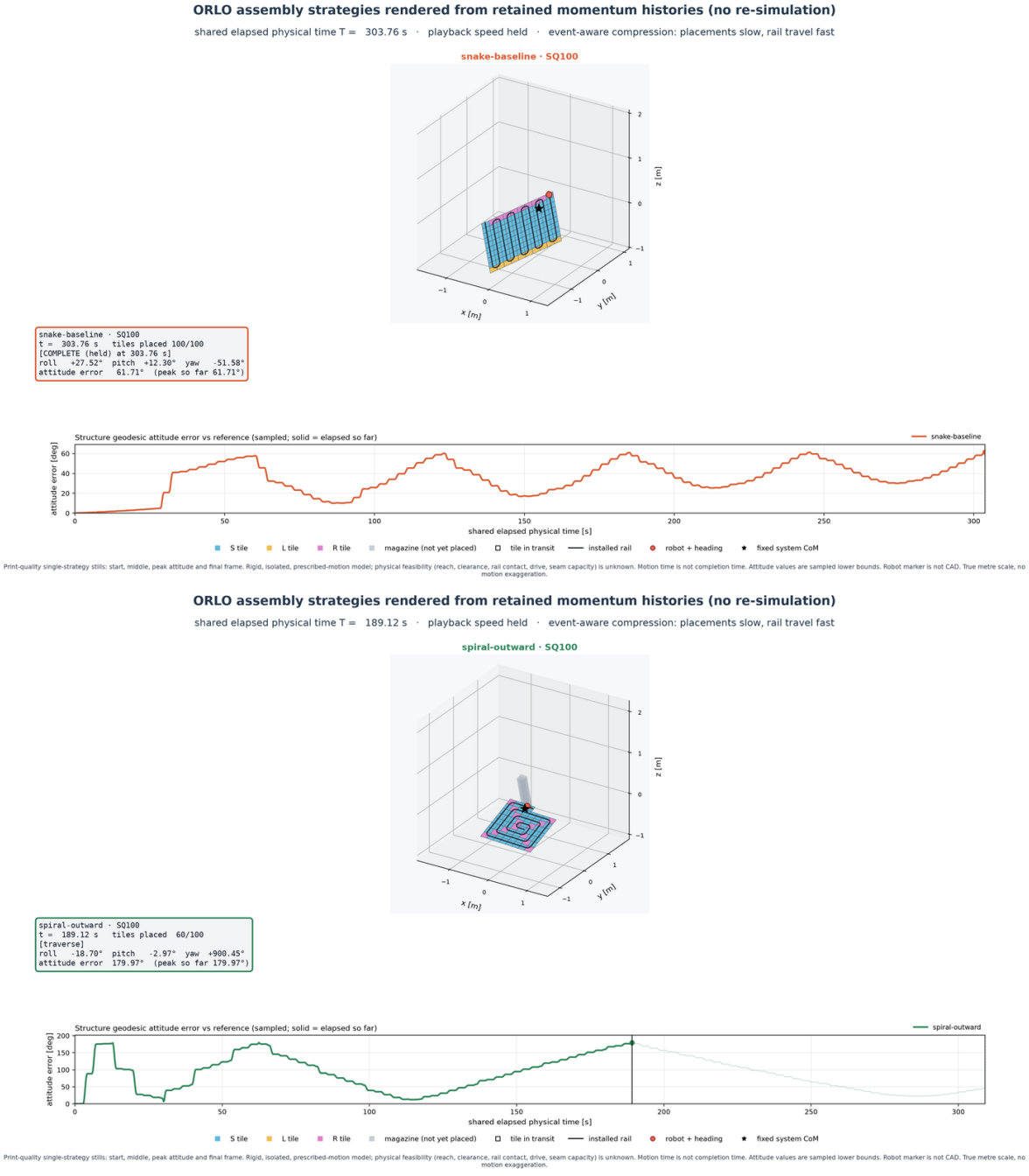


Figure 1. The structure at its sampled attitude peak. Top: snake-baseline, which tilts by about 62 degrees. Bottom: outward spiral, which sits at the limit of about 180 degrees. The robot marker is not CAD. Source: animation stills, rendered from the retained histories.

2 Method in brief

1. **Model.** A variable-mass rigid multibody in six degrees of freedom, with zero total linear and angular momentum. Robot, gantry transfers and tile seating follow prescribed smooth motions. The solver integrates the structure's attitude and origin with tight tolerances.
2. **Candidates.** A route is a path through the grid cells, plus the order and timing of moves. Two footprints were studied: **SQ100**, the filled ten-by-ten square, and **SHAPE100**, any flat, edge-connected shape of 100 tiles. The two footprints are separate problems and are never ranked against each other.
3. **Metrics.** Peak attitude error, final attitude error and motion time are reported together. We do not combine them into one score. A candidate is "in the partial best set" when no other candidate in the same comparison is at least as good on all three and better on one. This is a screening tool, not proof of a best choice.
4. **Checks.** Every run must pass conservation checks, schedule checks and solver-refinement checks. All peaks are *sampled lower bounds*: they are the largest value found at the time samples, and the true peak can be larger. A *sampling check* refines the samples and tests whether the peak stops changing. If the check fails, even the sampled value is not stable. A pass does not resolve a peak that is pinned at the half-turn limit, because the value cannot rise above that limit (this is the case for the outward spiral).
5. **Gates.** Eight physical gates (reach, clearance, magazine stack, rail, drive, fuse, joint load, mass) are carried in every report. All eight are *unknown* because the hardware data do not exist. An unknown gate is never counted as a pass.
6. **Evidence tags.** [V] verified (reproducible from retained files; see limit L8 for the code version); [A] assumed (an input or threshold, not a measurement); [U] unverified or unknown.
7. **Traceability.** Every number in this report is read from a *claim ledger*. A script checks the report against the ledger and the ledger against the hashes of the source files. No simulation was re-run to make this report.

3 What was learned, phase by phase

The project ran as numbered work packages (ZO-1 to ZO-23). We group them into seven phases. For each phase we say what it established or ruled out.

Phase 1: building blocks (ZO-1 to ZO-4). One file now holds all physical parameters. A snake path generator labels each cell as a straight tile or a left or right turn tile. A rail model gives the robot pose along straights and quarter-circle turns. A motion planner limits speed in turns. *Established:* turns, not straights, set the speed limit, because the sideways force needed at turn speed exceeds the rail force budget (60 N [A]). *Ruled out:* running at full speed through the turns.

Phase 2: the physics core (ZO-5, ZO-6, ZO-7, ZO-10). A six-degree-of-freedom momentum model, a seating model for tile placement with magnetic preload, a gantry model that lifts tiles from the stack, and a set of hard validation gates. *Established:* the stack and its out-of-plane centre-of-mass shift matter and need a three-dimensional model. A preload changes loads, not motion, so it is not attitude control. *Ruled out:* a two-dimensional model, and any force-driven transient claim that the corrected model does not support.

Phase 3: the first two strategies (ZO-8, ZO-9, ZO-11 to ZO-14). A1 and A2 were simulated, animated and compared. The first model kept the robot body fixed in orientation. A later version made the robot drive forward only and turn its whole body by a right angle at each turn tile. *Established:* the A1 and A2 trade is real. A1 turns the structure less (24.4° peak against 61.7° for A2) but takes 3.6 times as long (1086 s against 304 s). A1's cost includes 19 in-place pivots that are provisional [A]. *Ruled out:* calling either strategy an outright winner.

Phase 4: opening the design space (ZO-15 to ZO-18). The fixed snake was replaced by a general question: any route, any starting tile, any footprint of 100 tiles. We defined the metrics, audited the drive and rail limits, and built a programme interface for non-snake routes with strict rules (no travel over unbuilt tiles, no lost tile inventory). *Established:* the rules that make a comparison fair. *Ruled out:* optimising before the actuator and contact limits were written down.

Phase 5: candidates and first comparison (ZO-19 to ZO-21). We built a small library of routes: spirals, long runs, a staircase and the snake. We evaluated each with the same assumptions. *Established:* in SQ100 the

snake and the outward spiral form the partial best set. In SHAPE100 the straight chain and the staircase do. The inward spiral is excluded because its sampling check failed. *Ruled out:* a physical winner. The *physical* Pareto frontier (the best set when all hardware limits and the real build time are included) is empty, because joining time and joint demand are unknown.

Phase 6: a bounded search (ZO-22). A small search by "backbite" moves (re-wiring the end of the path) used 12 full solves and found 2 new routes. *Established:* those two routes do not beat the snake in their own comparison group. A timing change alone cannot reduce the peak or final attitude: the path in attitude space does not change when time is stretched. *Ruled out:* a quick gain from timing alone. The search is a small, informed sample, not a global search (L5).

Phase 7: stress test and evidence review (ZO-23). We changed robot mass, tile mass, inertia and centre-of-mass height, one at a time, and in two combined corners. *Established:* the snake keeps its ranking and passes all model checks. The spiral's peak stays at the limit and 8 of its 10 variants have unknown results. We also listed what the model cannot see: load peaks, power, magazine indexing, recovery from faults, and four deferred models (flexible structure, seam mating, thermal and power, external attitude control), each with its own "triggers" that tell us when that model is needed. *Ruled out:* reading the nominal ranking as robust beyond these two routes and these ranges.

4 Results

4.1 Attitude over time, SQ100

Figure 2 shows the attitude error over the build for four SQ100 routes. The snake rises to about 62° and ends there. The two spirals reach the 180° limit. The inward spiral is drawn dashed because its sampling check failed: the peak changed by 3.8e-05 rad between two successive sampling refinements, which is above the limit of 1e-6 rad.

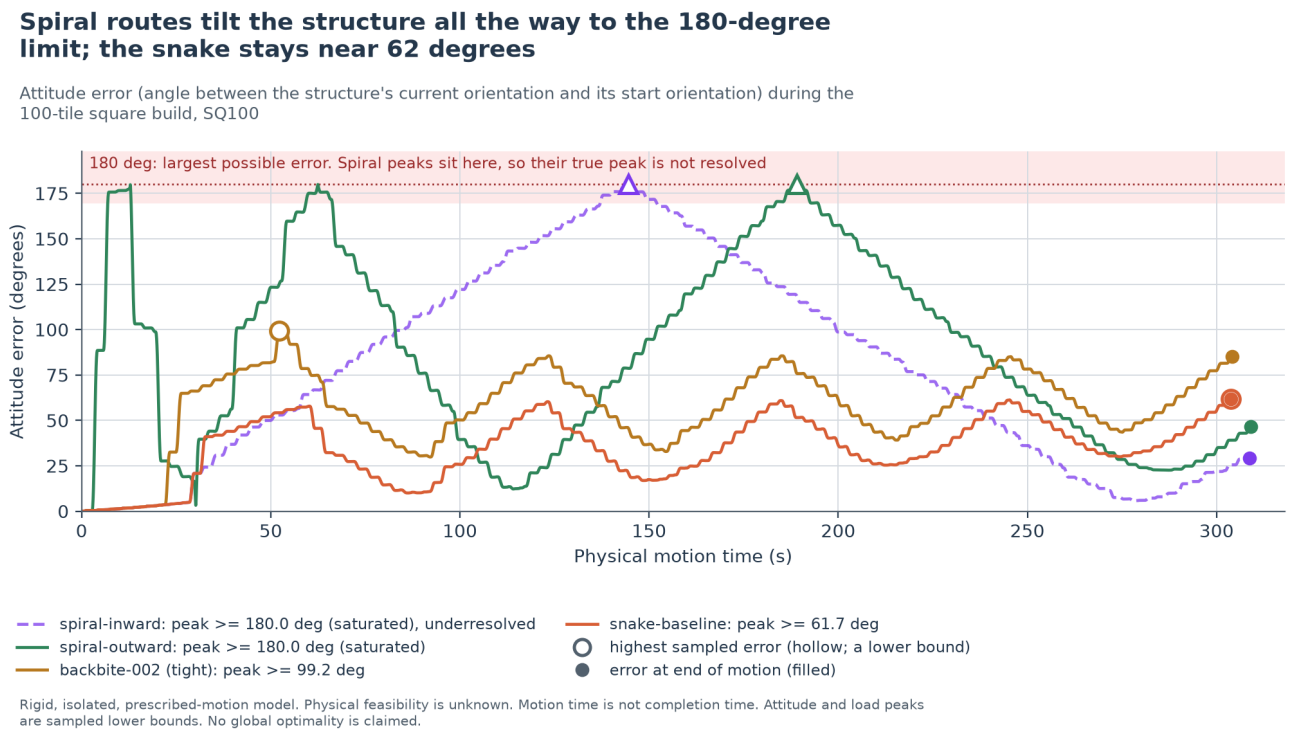
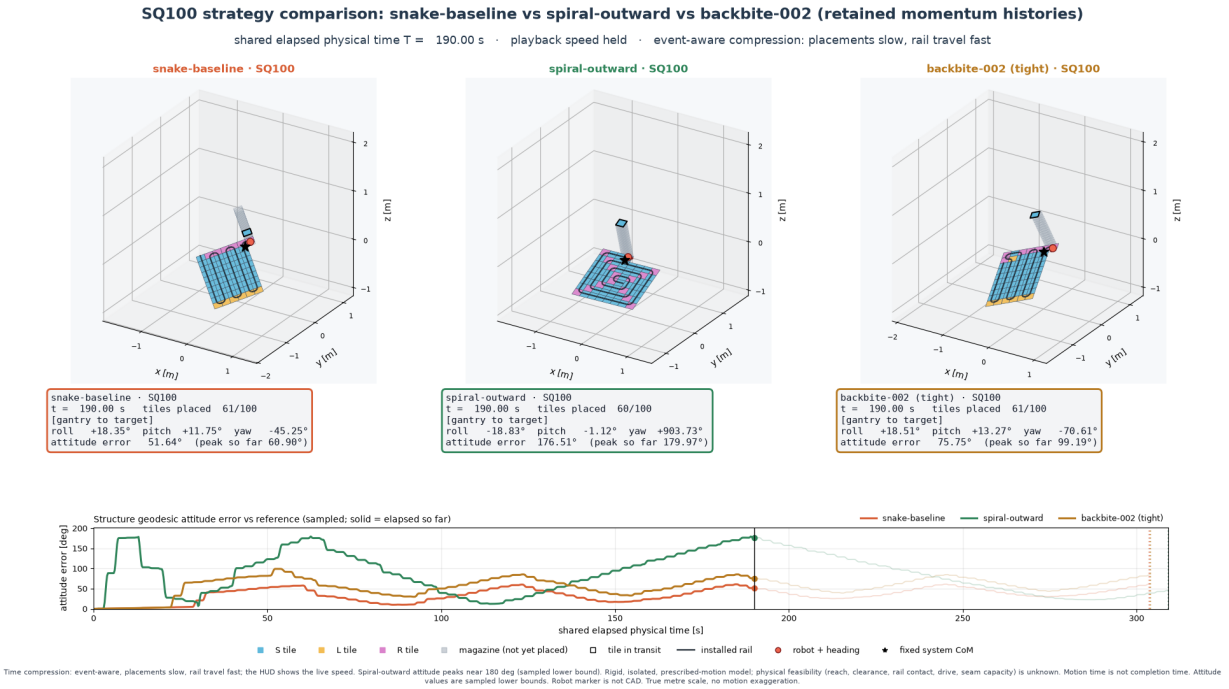


Figure 2. Attitude error against motion time for four SQ100 routes. The spiral peaks are pinned at the largest possible value, so the true peaks are not resolved. Tags: verified, with claim limits L1, L2 and L4.

Animation. The movie below plays the three main routes side by side in true metre scale, with a live clock.

The robot marker shows position and heading and is not CAD.



Movies: main comparison, three routes · four routes including the inward spiral · one gantry transfer and placement, close-up.

4.2 The tradeoff, SQ100

Figure 3 and the table below give the numbers. Motion times differ by 5.3 s between the two members of the partial best set, which is about 1.8 % of the snake's time (derived). The time axis does not separate them. The snake has the lower peak. The outward spiral has the lower final error. No route wins all three.

SQ100 routes. Motion only, nominal masses. Backbite and no-final-travel values are converted from radians (derived). The backbite routes were compared with the snake inside the search's own comparison groups; the snake beats both there on all three numbers.

Route	Motion time (s)	Peak error (deg)	Final error (deg)	Sampling check on peak	Tag
snake-baseline	303.8	61.7	61.7	passed	[V]
spiral-outward	309.1	180.0 (at the limit; unresolved)	46.4	passed	[V]
spiral-inward	308.6	180.0 (underresolved)	29.2	failed	[V]
backbite-001	304.4	133.3	83.5	passed	[V]
backbite-002	304.1	99.2	85.1	passed	[V]
snake without final travel	302.4	61.3 (unresolved)	58.3	failed	[V]

Two simple routes that look good on paper did not work well. The no-final-travel variant saves 1.4 s of motion time by skipping the last rail move, but its peak could not be resolved. The best of 2 backbite routes has a peak of 99.2°, above the snake.

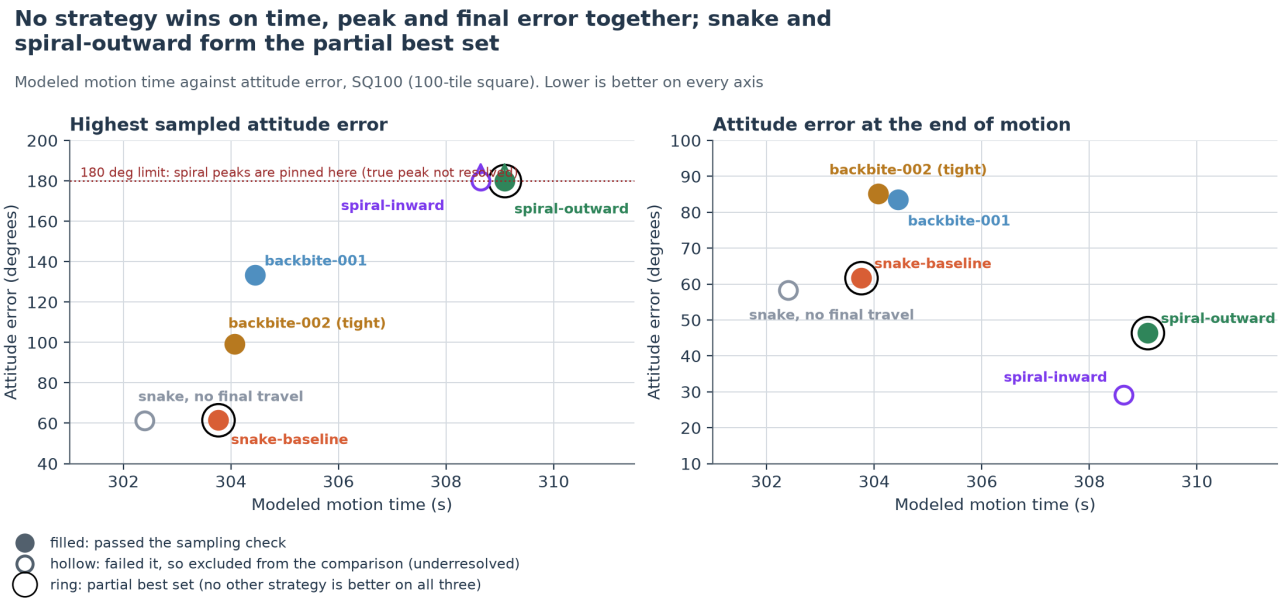


Figure 3. Motion time against peak error (left) and end-of-motion error (right) for six SQ100 routes. Rings mark the partial best set. Hollow points failed the sampling check. Tags: verified, with claim limits L1 to L5.

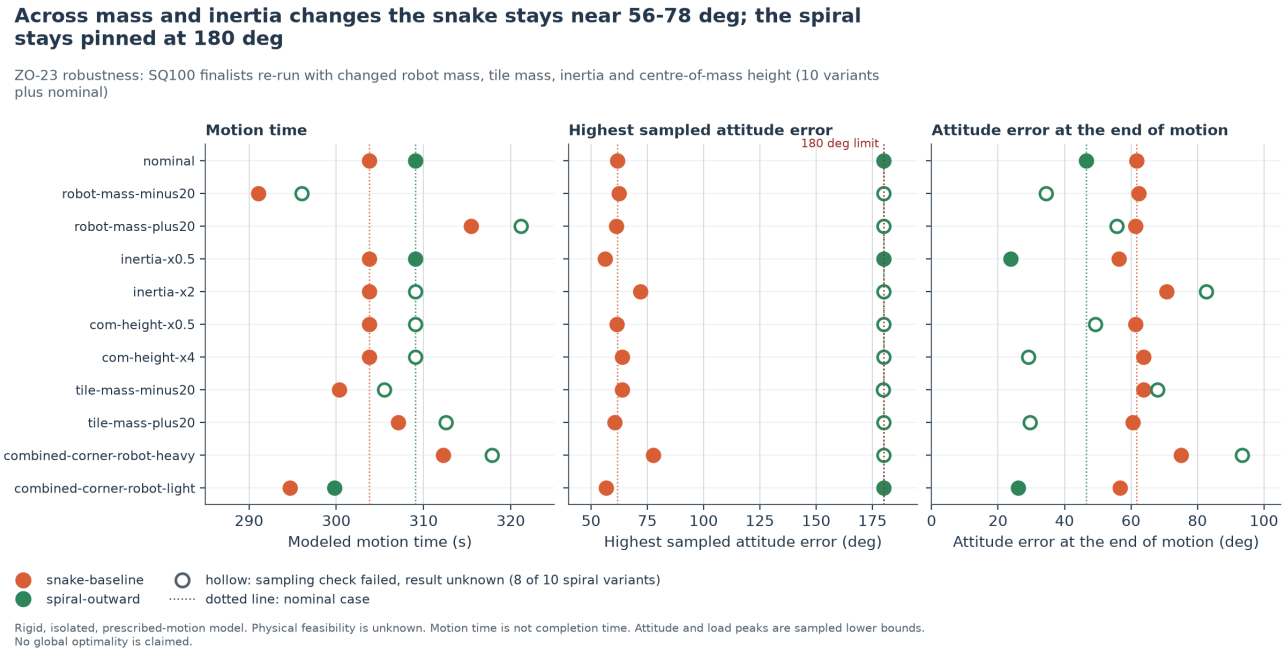


Figure 4. Motion time, peak error and end-of-motion error for the nominal case and ten changed variants (robot mass, tile mass, inertia, centre-of-mass height, two combined corners). Hollow points are unknown results. Tags: verified, with limits L1, L2 and L4.

4.3 Stress test of the two finalists

Across all variants the snake's peak stays between 56.3° and 77.6° and its final error between 56.3° and 75.1°. The worst case is the combined robot-heavy corner, where the peak is 25.8 % above nominal. Use that worst case as the working bound, not the nominal value. The spiral's peak stays between 179.8° and 180.0°, and its final error ranges from 23.7° to 93.4° (very sensitive to mass).

In 7 of 10 variants the nominal tradeoff holds. This is also arithmetic only: both routes pass all model checks (are *gate-eligible*) in only 2 of those variants, the ones where the spiral passes. In 3 the snake wins all three numbers, but this is arithmetic only, because the spiral result is unknown there. The number of rank changes where both routes are gate-eligible is 0. The snake is faster than the spiral in every case, by 4.9 to 8.1 s across all variants and drive deratings (a *derated* drive is a weaker drive; the schedule is re-planned for it). Limits: two routes only, input ranges that we assumed ([A]), and no study of how the changes interact.

4.4 A1 against A2

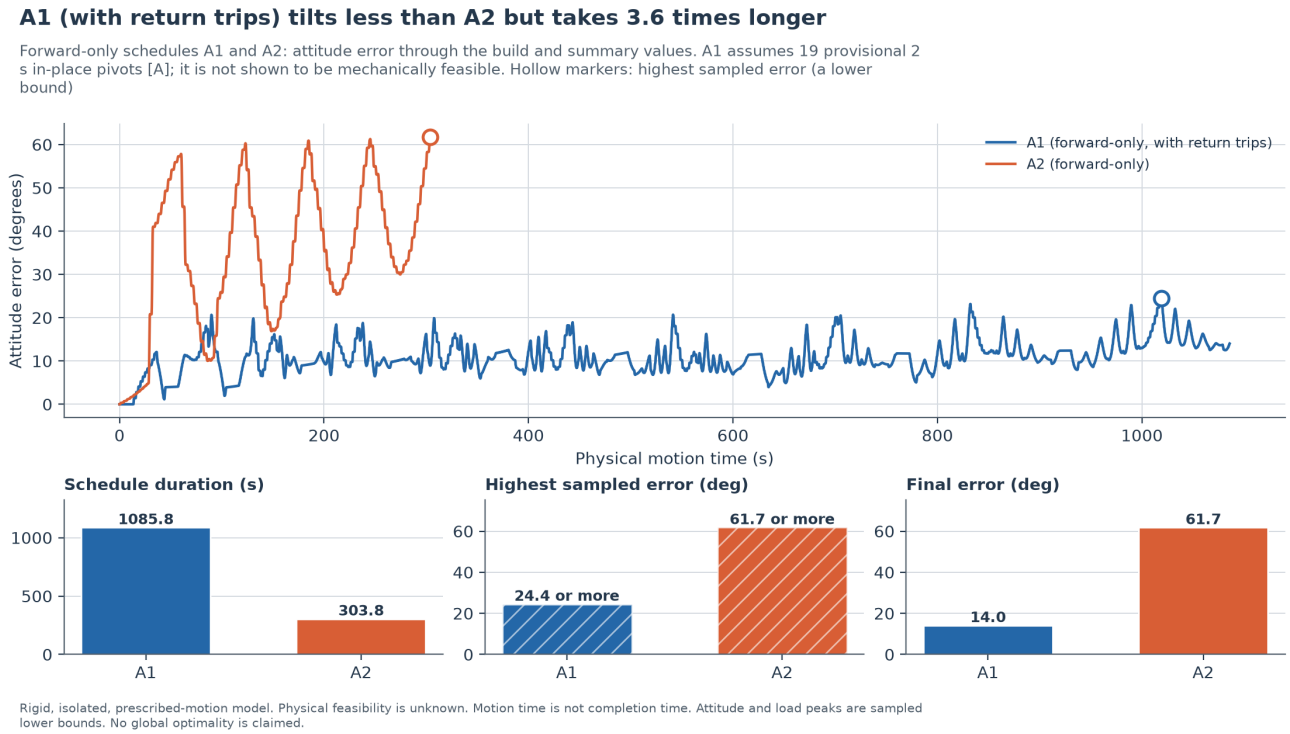
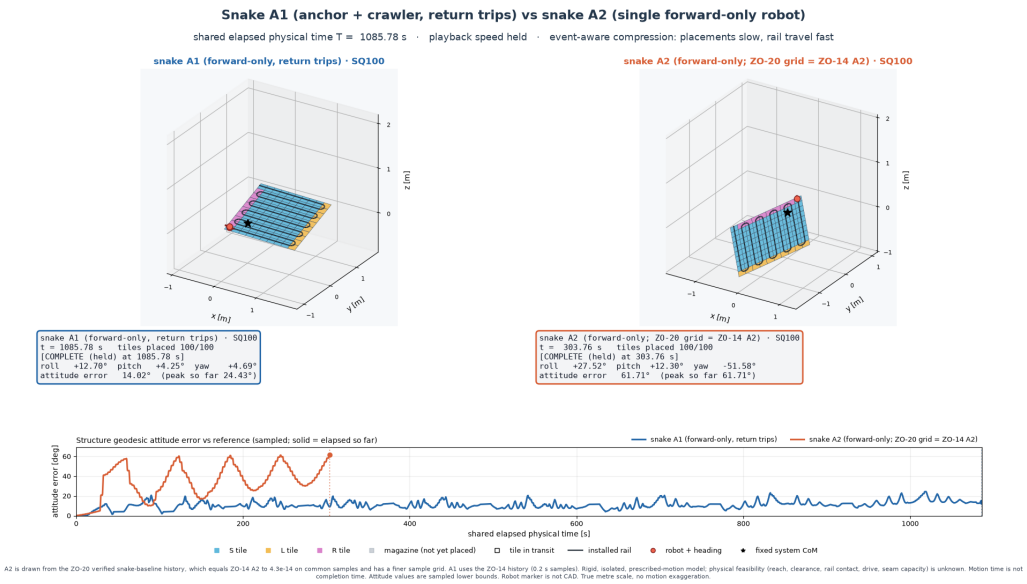


Figure 5. Attitude error over time for the forward-only A1 and A2 schedules (snake path). A1 tilts less but lasts much longer and needs provisional pivots. Tags: verified for model numbers, assumed for the pivots.

A1 reaches a peak of 24.4° and ends at 14.0°. A2 reaches 61.7° and ends there. A1 lasts 1086 s and A2 304 s. The A1 number includes 19 two-second in-place pivots that are assumed [A] and not shown to be mechanically possible. A2 is the same schedule as the snake-baseline.



4.5 Other footprints: SHAPE100

SHAPE100 is a different problem, so its numbers must not be read against SQ100 (L6). Within SHAPE100 the straight chain has the lowest peak (27.1°). The alternating staircase is the fastest (284.7 s, which is 16.4 s less than the straight chain) but tilts to 88.5°. The elbow (301.6 s, 52.5°) is beaten by the straight chain on all three numbers. The partial best set is the straight chain and the alternating staircase (straight-100 and alternating-staircase).

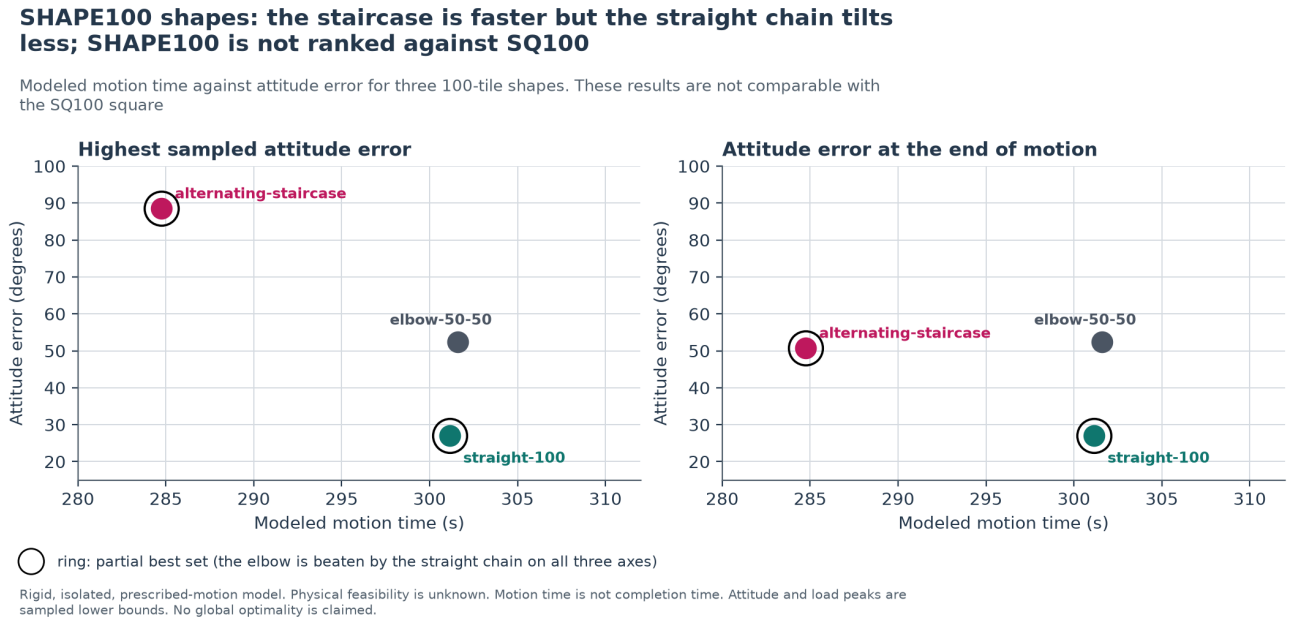
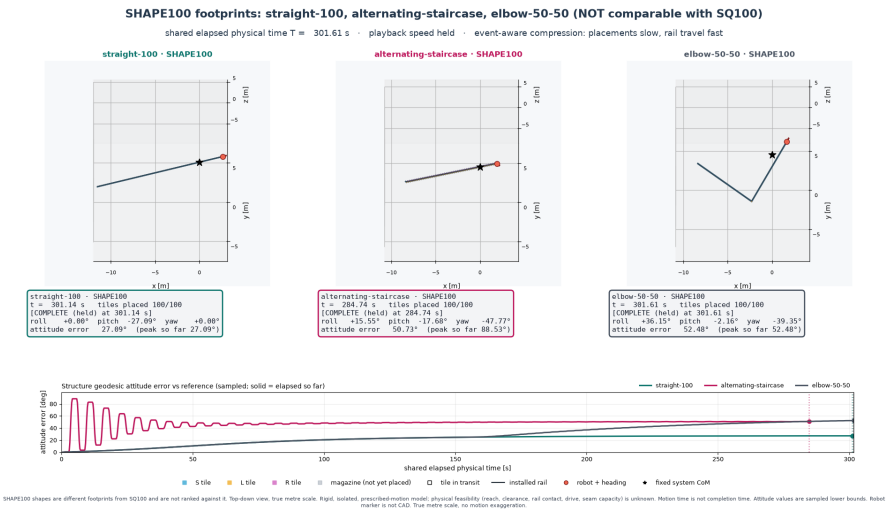


Figure 6. Same axes as Figure 3, for three SHAPE100 shapes. These footprints are different problems from the SQ100 square. Tags: verified, with limits L1 to L3 and L6.



4.6 Motion time is not completion time

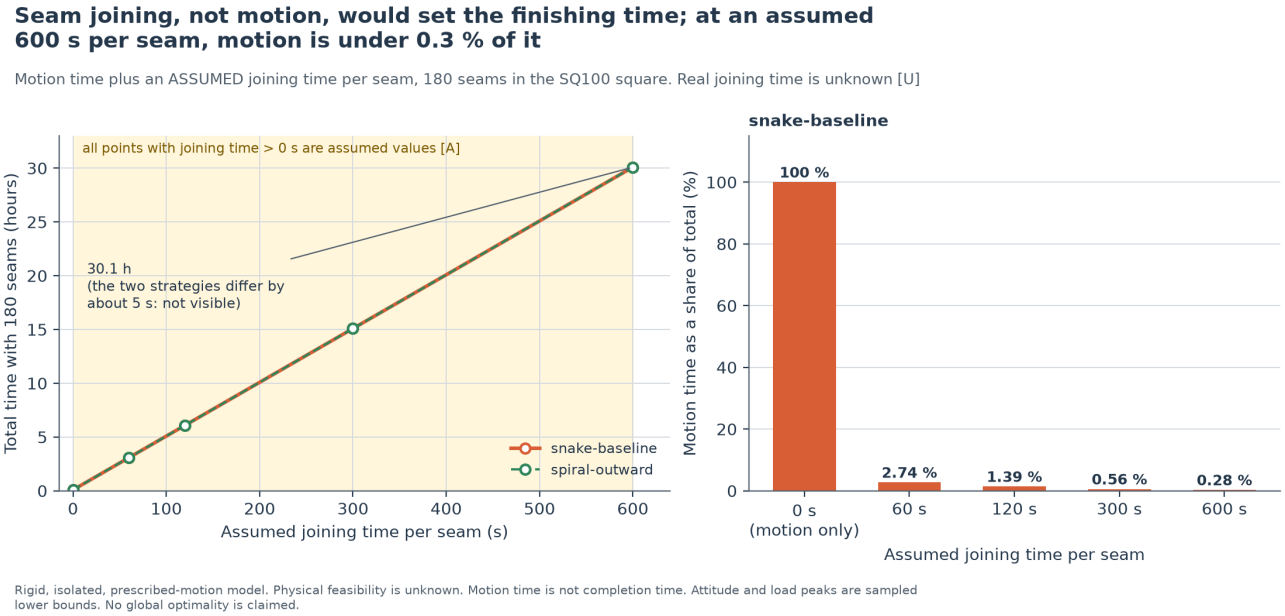


Figure 7. Total time for 180 seams at assumed joining times per seam. The joining times are assumptions; the real value is unknown, and is null in the reports, not zero. Limits L1, L2 and L3.

The model assumes that every seam is fused before the next placement ([A]). Each SQ100 route has 180 seams. How long one seam takes is unknown ([U]). Completion time is therefore the motion time plus 180 times the unknown joining and cooling time, plus any extra handling. At an assumed 600 s per seam the total for the snake is 108304 s, or 30.1 h. The motion is then 0.28 % of the total (assumed [A], derived). The two finalists differ by about 5.3 s, which does not show in a plot of hours.

4.7 Loads and power

The highest sampled force is 71.0 N and the highest sampled power is 570 W. Both occur at the first-tile gantry lift in every case. The closed-form value for that lift is 73.2 N and 646 W, so the sampled values are lower bounds. The only specified actuator has a nominal shaft power of 50.3 W ([A]). No gantry actuator is specified. The “gantry power” trigger (T2) is therefore tripped: the lift needs far more power than the one actuator we

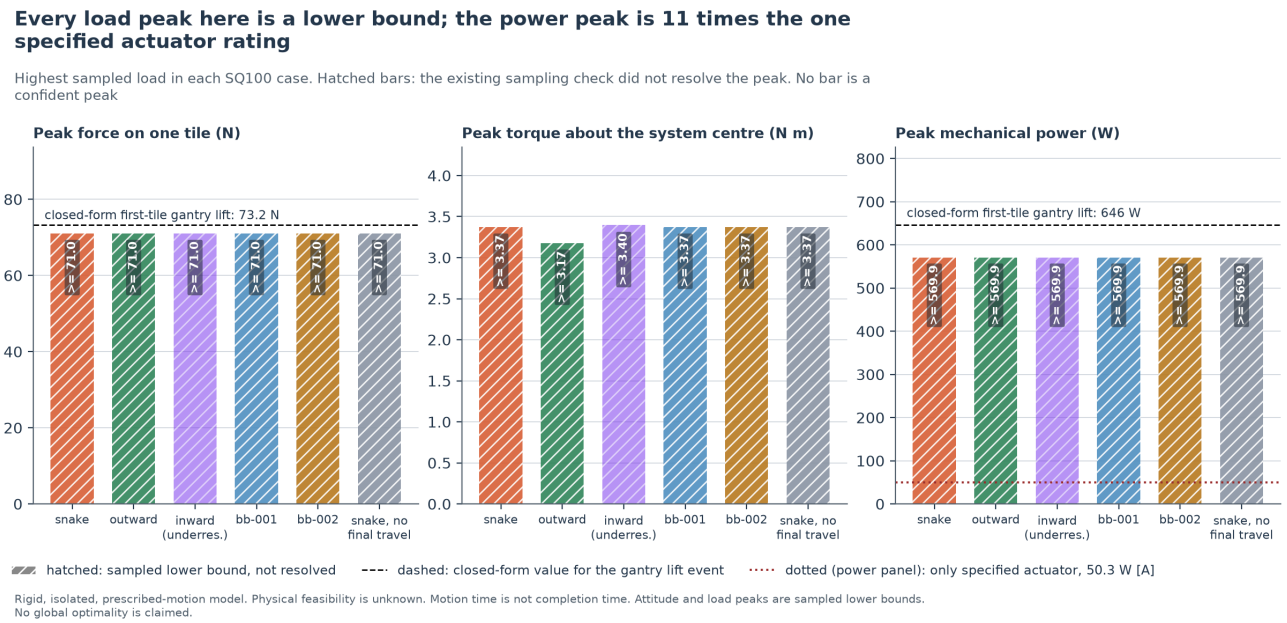


Figure 8. Highest sampled force, torque and power per case. All bars are hatched because the existing sampling check did not resolve any load peak. Tags: verified for sampled values, assumed for the actuator rating.

know. The peak torque on the centre of mass is 3.37 N m for the snake and 3.17 N m for the outward spiral. These are component resultants from a rigid model, not seam or actuator ratings.

To hold the structure fixed in space with an external attitude system, one would need at least 17.8 N m of torque and 7.33 N m s of stored momentum for the snake. No such actuator is specified and no pointing allowance exists.

4.8 Operational complexity

The snake needs 3 distinct launch orientations in the magazine and 6 distinct tile-and-orientation states. The outward spiral needs 4 and 8. Next to each other in the stack, 27 tile pairs of the snake and 34 of the spiral need different states. The spiral also needs 116 rest-to-rest rail traverses against 108 for the snake. Whether a mixed-orientation column of 2.3 m can survive launch and unloading is unknown ([U]). Fault detection, abort and restart are not modelled ([U]).

5 Decision rationale

The evidence supports a *reference*, not a choice.

1. **Carry the snake forward as the SQ100 reference.** It has the lower peak, fewer magazine states, fewer traverses and the cleanest robustness record. This is a model result and says nothing about hardware.
2. **Keep the outward spiral as a conditional alternative.** It is attractive only if a low final error matters more than a peak at the limit. Its unknown results need a refined solve first. Do not promote it on final error alone.
3. **Do not promote the backbite routes, the inward spiral or the no-final-travel snake.** None beat the snake in its own comparison group, or the sampling check failed. This does not prove they are worse on hardware.
4. **Do not rank SHAPE100 against SQ100.** A free footprint is a separate decision that needs its own evidence.
5. **Get physical evidence before committing.** The choice between snake and spiral may be settled by reach, magazine or rail limits, which the model cannot see.

6 Next validation experiment

Three experiments were considered. Each targets a gap that the model cannot close.

Experiment	Question	What would change the decision
E1. Seam joining on a bench coupon (small effort)	How long does one seam take until it carries load, and can it overlap with robot motion?	If joining is long and cannot overlap, motion-time differences stop mattering and the choice rests on attitude, complexity and joint load. A joint that cannot carry the loads removes routes that load two seams at once.
E2. Gantry reach, clearance and magazine mock-up (medium effort)	Can the gantry take each tile from a tall mixed-orientation stack and place it in the modelled time, and does the stack keep its order through launch?	If the stack cannot hold the orientations the spiral needs, the spiral loses first. If the gantry cannot meet the lift profile, the schedule and all loads change.
E3. Rail pinion contact and turning test (medium to large effort)	Can the drive hold the planned speed and turn on the rail with real masses?	If turns cannot run at the planned speed, motion times change and routes with many turns or stops lose ground.

Suggested order (a recommendation, not a decision): E1 and E2 first. E1 is small and tells us whether motion time matters at all. E2 can remove a whole route family and covers the one tripped trigger (T2). E3 follows E2, because the loaded mass and stop pattern come from the E2 design. The project owner will record the final choice.

Glossary

A1, A2. Two work organisations: A1 uses an anchor and a crawler with return trips; A2 uses one mobile robot that carries all tiles.

Angular momentum. The rotational counterpart of momentum. For an isolated system it is conserved.

Attitude, attitude error. The orientation of the structure. The error is the smallest rotation angle between the current and start orientations (zero to half a turn).

Backbite route. A route made by re-wiring one end of an existing path.

Comparison group. A set of runs made with the same solver and sampling settings. Routes are only compared inside one group. The search routes were run with tighter settings than the route library, so they were compared with a snake run at the same tighter settings.

Derated drive. A drive with less force or speed than nominal. The schedule is re-planned for it.

Centre of mass. The mass-weighted mean position of the whole system.

Claim ledger. The table of every headline number with its source file and evidence tag. Every number here comes from it.

Completion time. The time to finish the build, including seam joining. Unknown.

Free-floating. With no external force or torque. Momentum is conserved.

Gantry. The mechanism that lifts a tile off the stack and puts it into place.

Gate. A pass-or-fail check. Unknown is never counted as a pass.

Gate-eligible. A run that passes all model checks, so it can take part in a ranking.

Magazine. The tall stack that holds the tiles that are not yet placed.

Motion time. The modelled time for robot motion, gantry transfers and tile seating. It does not include seam joining.

Partial best set. The routes that no other route in the same comparison beats on all of motion time, peak error and final error.

Pivot. A turn in place. Used by A1 on its return trips (assumed).

Rail. The track on the tiles on which the robot drives. The robot drives along it with a gear (pinion) on a toothed track; *rail pinion contact* is the contact between them.

Sampled lower bound. A peak found from a finite set of time samples. The true peak is at least this large.

Seam. The edge where two tiles meet. Each seam must be joined.

Snake route. Rows traversed back and forth, the standard route.

Unwrapped yaw. The total turn about the vertical axis, counted without folding it back into the range of plus or minus half a turn. Two full turns count as two full turns, not as zero.

SQ100, SHAPE100. The filled ten-by-ten footprint, and the free flat shapes of 100 tiles.

Spiral route. A path that winds in (inward) or out (outward) between the centre and the border.

Trigger (T2, M3, A1, F1). A condition that tells us when a deferred model is needed. The robustness study lists several triggers for each of four deferred models; this report names four of them. T2 is gantry power, M3 placement load, A1 an attitude allowance, F1 structure flexibility. (This A1 is not the strategy A1.)

Known, assumed and unknown

The list below follows the claim ledger. The last figure shows the same split.

Known, within the model [V]

- The motion times, peak and final attitude errors in the results tables, and the partial best sets.
- The snake passed all model gates in 10 of 10 variants. The spiral's result is unknown in 8.
- A1 tilts less than A2 and takes 3.6 times as long.
- Timing changes alone cannot change peak or final attitude.
- The highest sampled force, torque and power values, and the closed-form gantry values.
- The gantry power trigger (T2) is tripped. The placement load trigger (M3) is not.
- The seam count: 180 in SQ100, 99 in each SHAPE100 shape studied.

Assumed [A]

- 100 tiles of 0.5 kg, a dry robot of 85 kg, a system of 135 kg.
- Tile pitch 0.160221 m. Magazine height 2.2955 m (ideal, pre-indexed, top access).
- A rail force budget of 60 N, gantry lift, traverse and lower times of 0.3, 0.6 and 0.3 s, a seating preload of 50 N and a seating time of 0.3 s.
- The system starts at rest with zero momentum, isolated, with no external attitude control.
- The A1 pivots (19 of them, two seconds each).
- The seam joining time of 600 s used only to illustrate (L3).
- The stress ranges for mass, inertia and centre of mass.

Unknown [U]

- Gantry reach, clearance, magazine indexing, rail contact, drive, seam capacity and joint safety: all eight gates are unknown.
- The joining time per seam, and therefore the real completion time.
- Electrical energy and power, joint load, drive load.
- Whether any pointing or attitude allowance exists.
- Whether a mixed-orientation column survives launch and unloading; fault recovery.
- Structure flexibility (the model is rigid).
- Whether any route is the global best (L5).

Where to find everything

- Technical decision package for the project owner: `deliverables/zo24/decision_package/README.md`.
- Claim ledger and source hashes: `deliverables/zo24/evidence/`.
- Figures with captions: `deliverables/zo24/figures/figures.md`. Movies, stills and render manifest: `deliverables/zo24/animations/`.
- Robustness report: `deliverables/robustness/README.md`.
- Rebuild this report with `bash deliverables/zo24/report/make_report.sh`.

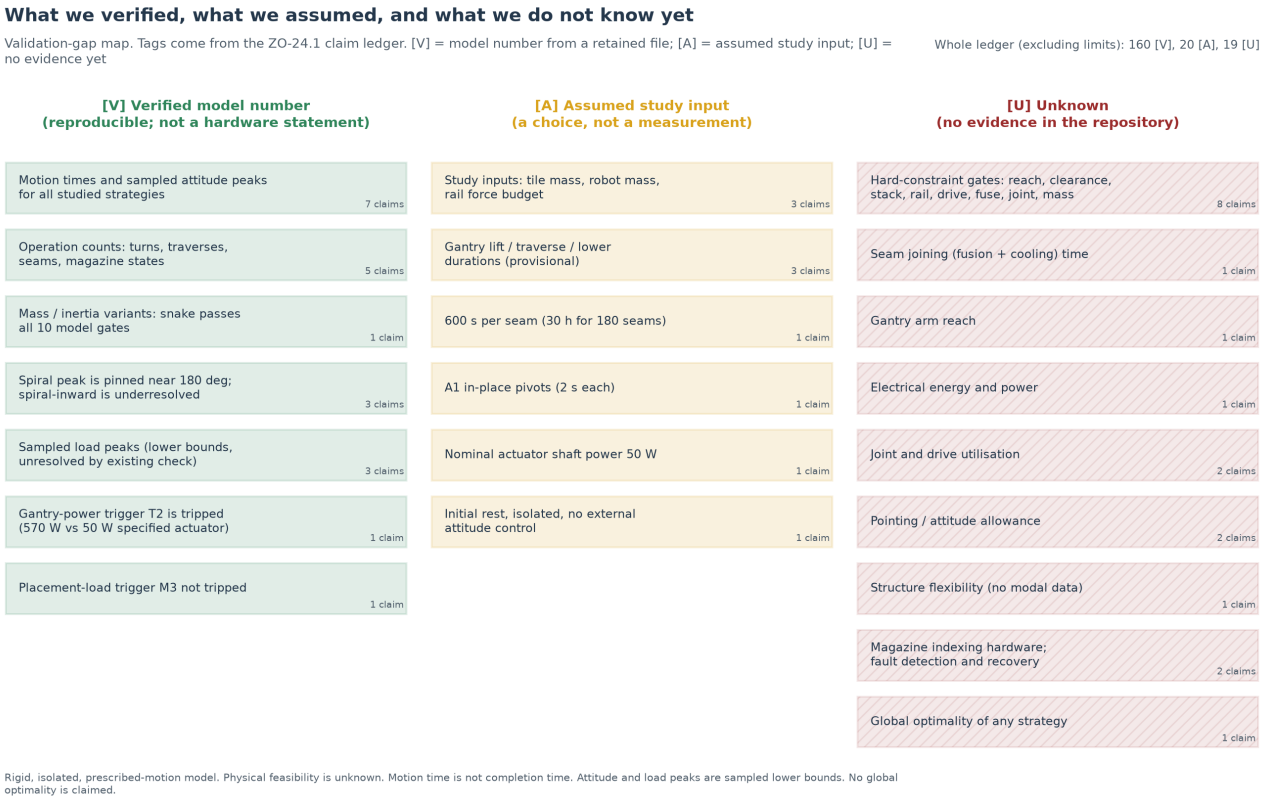


Figure 9. What was verified, what was assumed and what is unknown. Hatched items are unknown.